Predicting Port Congestion and Ship Delays

1. Problem statement

Ships queue for berths. When all of a port's berths are busy, arriving ships wait at anchor, and that wait is expensive: the ship sits idle, the cargo arrives late, and trucks and warehouses booked for it go unused. This happens at busy ports everywhere, and without a forecast a long wait usually becomes clear only when it is too late to re-plan.

The task: for every arriving ship, predict the **likelihood of delay**, give a **confidence score**, and **explain the key factors**, early enough for planners to act.

2. Our approach

We built a system that predicts whether a cargo ship arriving at a Brazilian port will wait **more than 24 hours** before it gets a berth. For every ship it gives four things: the **chance of a delay**, a **confidence score**, the **expected wait in hours** with a likely range, and the **main reasons** in plain words. It can predict when the ship arrives, or **3 days earlier**, while there is still time to act.

It is trained on **85,606 real port calls** at Brazil's 15 busiest ports (2020–2024) from ANTAQ, the national waterway regulator, plus **daily weather at each port** (rainfall, maximum wind speed and wind gusts) from the free **Open-Meteo historical weather archive**.

A demo app shows each prediction on a 3D view of the port, explains it, and lets a planner test "what if" changes such as more ships arriving or a berth closing.

What the system answers. For a cargo ship arriving at one of Brazil's 15 busiest ports, it answers four questions:

1. **Will it be delayed?** A ship is *delayed* if it waits **more than 24 hours** between arriving at the port and getting a berth. A one-day slip is where plans for trucks, warehouse slots and deliveries usually break. The cut-off is a setting and can be changed.
2. **How sure are we?** A **confidence score**: how likely the yes/no answer is to be right.
3. **Why?** The **main causes** in plain words: a long line of ships, busy berths, a terminal that has been slow lately, the ship's own track record, and so on.
4. **For how long?** The **expected wait in hours**, with a range the real wait falls inside **80%** of the time.

When it answers. At two moments, using only what is known at that time:

Moment	Who uses it	What they can do
When the ship arrives	Port and landside planners	Plan berths, trucks and warehouse slots
3 days before	The ship operator	Slow down to save fuel, warn the customer, or consider another port

How well it works. Tested on all **17,417 arrivals of 2024**, a year the system never saw while it was being built:

Measure	When the ship arrives	3 days before	What it means, in one line
Accuracy	**78.7%**	**75.1%**	Out of 100 ships, about 79 get the right yes/no answer ("delayed" or "on time").
Delays caught	**94.8%**	**93.8%**	Out of 100 ships that really waited more than 24 hours, the alarm went off for about 95.
ROC-AUC	**0.873**	**0.836**	Pick one delayed and one on-time ship at random: 87 times out of 100 the delayed one gets the higher risk.
Typical hours error	**17.6 h**	**20.2 h**	Half of all wait estimates are within about 18 hours of the real wait.

3. System architecture

The system is built on **four LightGBM models**. All four models use the same **108 features** and are **retrained on the 1st of every month** on every port call whose outcome is already known by that day.

1. Main model: LightGBM classifier (the delay model). Answers *"will this ship wait more than 24 hours?"* It outputs **p, the chance of a delay** between 0% and 100%. It uses **1,269 trees** for the at-arrival prediction (773 for 3 days before), each with up to 63 leaves, added with a small learning rate (0.01) so that each tree only makes a small correction. The number of trees was set by early stopping on held-out data. From p come the **yes/no answer** (delayed if $p \geq 23\%$), the **confidence score** ($\max(p, 1 - p)$), and the **reasons** (each feature's share of p, from TreeSHAP).

2–4. Range models: three LightGBM models for the waiting time. These answer *"how long will it wait?"* Instead of one single guess, each of the three models estimates one bound of the wait:

- **Lower-bound model:** estimates the shortest likely wait. Only 1 ship in 10 waits less than this.
- **Expected-wait model:** estimates the middle wait. This is the expected wait shown to the planner.
- **Upper-bound model:** estimates the longest likely wait. Only 1 ship in 10 waits longer than this.

The lower and upper bounds together give the **expected waiting range**, for example *"expected wait 27 hours, likely between 4 hours and 3 days"*. We widen this range slightly so that the real wait falls inside it **80% of the time**; on 2024 it did so for 79.8% of ships.

4. Results

The results on the 2024 test set can be reproduced by running python src/show_results.py, which recomputes them from the saved predictions:

```
Windows PowerShell

PS C:\Port-congestion-predictor> python src/show_results.py

Test on all 2024 arrivals (never used to build the models)

                                 At arrival   3 days before
Ships tested                         17,417          17,417
Accuracy (yes/no right)               78.7%           75.1%
Delays caught (alarm line)            94.8%           93.8%
ROC-AUC                               0.873           0.836
Typical hours error (median)          17.6h           20.2h

Alarm line: 23% at arrival, 23% three days before
```

Figure 1. Output of the results script on the 2024 test set.

Figures 2 and 3 show one real case: a container ship arriving at the TCP terminal in Paranaguá on 1 January 2024. The system gave a 55% chance of delay; the ship really waited 46 hours.

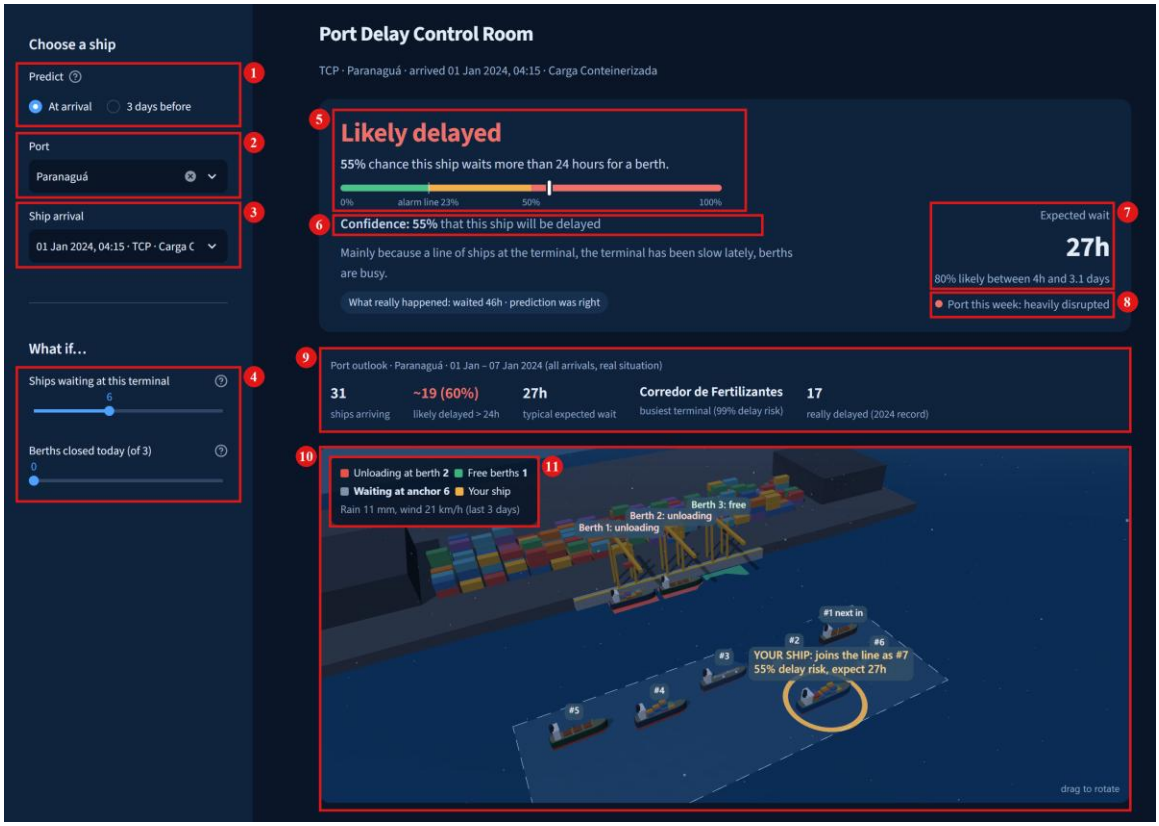


Figure 2. Choosing a ship and reading its prediction.

1. **Prediction moment:** predict when the ship arrives, or 3 days before.

2. **Port:** any of the **15 busiest Brazilian ports**.
3. **Ship:** any real 2024 arrival at that port (2,282 at Paranaguá).
4. **What-if sliders:** add ships to the line at this terminal, or close some of its berths (this terminal has **3 berths**); every number on the page updates.
5. **Verdict:** the chance of waiting more than 24 hours (55%), on a gauge that marks the 23% alarm line.
6. **Confidence:** how likely the yes/no answer is to be right.
7. **Expected wait:** 27 hours, and the range the real wait falls in 80% of the time (4 hours to 3.1 days).
8. **Port status this week:** normal, slightly disrupted or heavily disrupted, from the port's average wait over the last 7 days.
9. **Port outlook:** for the whole port this week, the ships arriving, how many are likely to be delayed, the typical wait and the busiest terminal.
10. **3D view of the terminal:** its berths, the ships unloading, the ships waiting at anchor in order, and this ship highlighted.
11. **Live counts:** ships unloading, free berths, ships waiting at anchor, and the last 3 days of weather.



Figure 3. Why the ship is likely to be delayed, and how today differs from a normal day.

12. **What led to the delay:** the main causes, each with how many percentage points it adds to or removes from the chance of delay. Here a line of ships (+18 points) and a terminal that has been slow lately (+7) drive the risk.
13. **What to do:** suggested actions for planners, based on the risk and the expected wait.
14. **Today vs a normal day:** today's situation next to this terminal's normal (its typical value in 2020–2022). Here 6 ships are waiting where 1 is normal, and recent waits are 41 hours against a normal 12.